



EXPERIMENT NO-5

Study and construction of Half Subtractor using NAND logic gates.



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Experiment No – 5, Study and construct a half subtractor using NAND Universal digital logic gates.

Theory:

A half-subtractor serves as an essential arithmetic component designed for the subtraction of two binary digits. This circuit incorporates two input ports, representing the two binary bits to be subtracted, and delivers a pair of outputs: one for the resulting DIFFERENCE (DIFF) and the other for the BORROW generated during the addition process.

Figure 5.1 shows the block diagram of a half subtractor and Table 5.1 shows the truth table of a half-adder, showing all possible input combinations and the corresponding outputs.

The Boolean expressions for the DIFFERENCE and BORROW outputs are given by the equations,

Let us denote the **Difference as (D)** and **Borrow as (Br)**.

$$D = \bar{A}.B + A.\bar{B} \dots\dots\dots (5.1)$$

$$Br = A.\bar{B} \dots\dots\dots (5.2)$$

To construct the half adder using NAND logic gates we need to modify the equation no 5.1 as given below;

$$D = \bar{A}.B + A.\bar{B}$$

$$D = A.\bar{A} + \bar{A}.B + A.\bar{B} + B.\bar{B}, \text{ introduce two additional terms which is equal to logic } - 0.$$

$$D = A.(\bar{A} + B) + B.(\bar{A} + \bar{B}), \text{ the common terms.}$$

$$D = \overline{A.(\bar{A} + B)} + \overline{B.(\bar{A} + \bar{B})}, \text{ by applying double negation}$$

$$D = \overline{A.(\bar{A} + B)} + \overline{B.(\bar{A} + \bar{B})}, \text{ applying De-Morgans theorem.}$$

$$D = \overline{A.(\bar{A} + B)} + \overline{B.(\bar{A} + \bar{B})}, \text{ again applying De-Morgans theorem.} \dots\dots\dots (5.3)$$

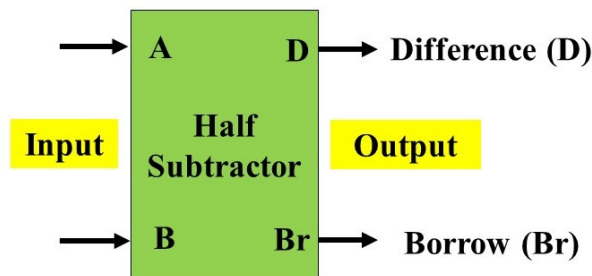


Figure (5.1), block diagram of half subtractor.

B	A	DIFFERENCE (D)	BORROW (Br)
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Figure (5.1), Truth Table of half subtractor.

As evident from the truth table the **DIFFERENCE (D)** can be implemented using a XOR logical gate and **BORROW (Br)** can be implemented using two inputs AND logic gate. But to implement the half

subtractor using NAND gates we need to implement the equation (5.3). The figure (5.2) shows the logical circuits using NAND logic gate.

To implement the BORROW (Br), we just need to invert the output $\overline{\overline{A.(A.B)}}$ as shown below;

$$\text{Let } M = \overline{\overline{A.(A.B)}}$$

$$\overline{M} = \overline{A.(A.B)}$$

$$= A . (\overline{A + B})$$

$$= A . \overline{A} + A . \overline{B}$$

$$= \overline{B} . A \quad (\text{as } A . \overline{A} = 0)$$

(B.A) of the first NAND logic gate.

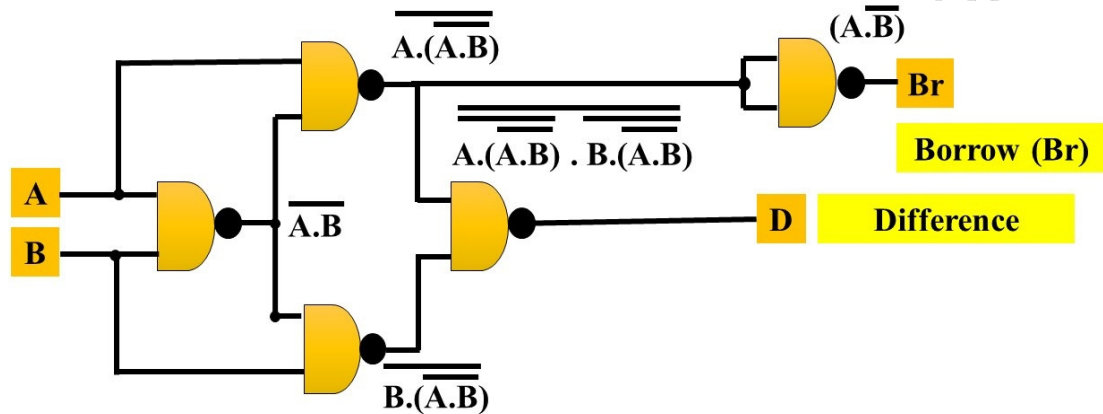


Figure (5.2), Logical circuit diagram using NAND logic gate.

Semiconductors and Instruments required

1. DC +5V regulated power supply.
2. Wish/Bread board.
3. Jumper wires.
4. Digital voltmeter.
5. Quad 2-input NAND gate (7400) – 2 nos

Implementation and Circuit diagram

The circuit can be implemented using a two NAND (7400) logic IC.

The circuit given in the figure (5.3) shows the actual implementation of half adder using NAND gates.

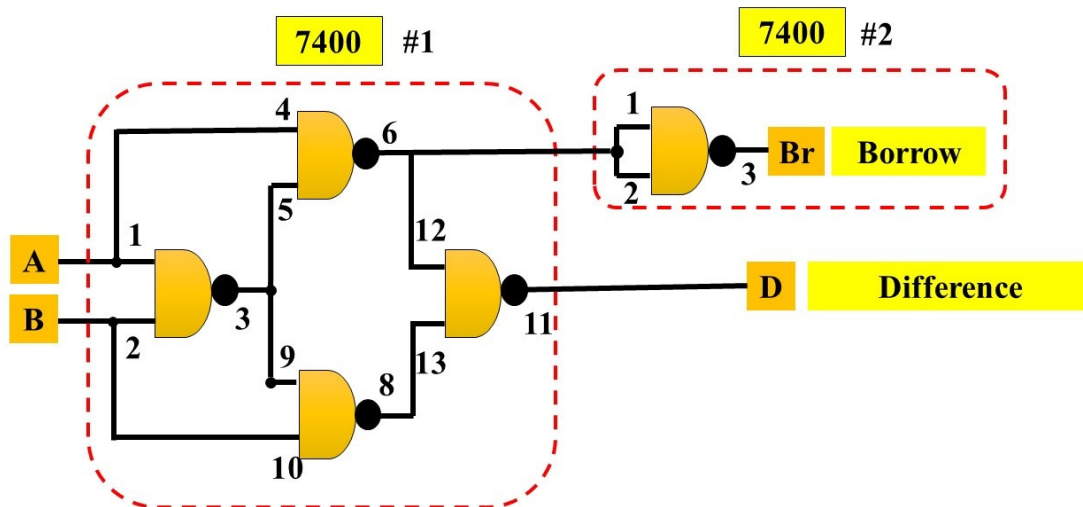


Figure (5.3), Implementation circuit diagram using NAND logic gate.

Result/Observation

Truth table for the half adder using NAND logic gates.

Logical Input		Logical output		Output voltage	
B	A	DIFFERENCE (D)	Borrow (Br)	DIFFERENCE (D)	Borrow (Br)
0	0	0	0		
0	1	1	0		
1	0	1	0		
1	1	0	1		

Table 5.2, Experimental observation.

Discussion

1. The power supply used must be regulated to +5V dc.
2. The jumpers used on the bread board must be short.
3. All the IC used must be properly seated on the bread board along with proper connection with regulated power supply.
4. The inputs to be connected to logic - 0 must not be left floating. For logic-1 and logic-0 we need to connect the inputs either to high voltage i.e. **+5V (V_{CC})** or to the ground potential.
5. Outputs can be visualized by connecting LED. Those must be connected via current limiting resistor.